

CPU. The die size is affected by the fabrication process and the number of transistors included within.

1.3.6 Fabrication process

This term is used to refer to the smallest distance between two components in the CPU. All CPUs contain miniature circuits etched on a silicon chip. The latest Intel CPUs use a 45-nanometre process.

1.3.7 FPU

Floating Point Unit; a sub-unit of the CPU core. The CPU is made up of different functional sub-units. There is the ALU (Arithmetic and Logical Unit), the FPU, Registers (temporary storage areas), and the caches. The CPU Core refers to all components of the CPU except the caches; it forms the area that is responsible for the actual computation.

1.3.8 FSB

The Front Side Bus is the data channel between the CPU and the system RAM. In turn, the Back Side Bus refers to the link between the CPU and the cache memory within the CPU die itself. The FSB frequency is the speed at which the bus operates, and is an indicator of the number of times the CPU interacts with the rest of the system. Usually, the CPU itself works much faster than the FSB (see Multiplier).

1.3.9 LGA

A Land Grid Array package is the latest form of packaging, seen in Intel's latest CPUs. This form of package does away with pins or other projections, and offers sockets only. The pins are present on the socket in the motherboard.

1.3.10 Multi-core CPUs

Many present CPUs are multi-core, meaning that there exist more than one computational unit on the same CPU. This allows for better performance when dealing with several jobs simultaneously. Both the major manufacturers offer quad-core CPUs.